

5D-to-4D Gauss-Codazzi Zero-Mode Projection and Sakharov-Abram Induced Gravity in Geotopological Hydrodynamics

Taquan Abram

Professional Engineer, Flint, Michigan, USA

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ABSTRACT

We formalize the dimensional reduction and gravitational induction mechanism in Geotopological Hydrodynamics (GTH v12.0), proving how four-dimensional Einstein-Hilbert gravity and effective metric field equations emerge from zero-mode projections of the 5D substrate continuum. We execute the 4+1 Gauss-Codazzi decomposition of the 5D Einstein field equation $G_{AB}^{(5)} + \Lambda_5 g_{AB} = \kappa_5 T_{AB}^{(5)}$, deriving the effective 4D boundary equation $G_{\mu\nu}^{(4)} + \Lambda_{\text{curv}} h_{\mu\nu} = \kappa_{\text{SI}} T_{\mu\nu}^{\text{eff}} + C_{\mu\nu}$, where $C_{\mu\nu}$ is the projected Gauss-Codazzi residual tensor incorporating Weyl electric projections $E_{\mu\nu} = C_{ABCD} n^A e^B_{\mu} n^C e^D_{\nu}$ and extrinsic curvature strains $K_{\mu\nu}$. We evaluate the Schwinger proper-time one-loop fluctuation determinant $W_{\text{eff}}[g] = -\frac{1}{2} \int_0^{\infty} (ds/s) e^{-M_0^2 s} \int d^4x \sqrt{-g^{(4)}} K_E(x,x;s)$, proving the Sakharov-Abram emergent coupling formula $G_{\text{model}} = 3\pi \kappa_s / (4 M_{\text{UV}}^2) = 6.5406 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ under state vector $\Theta = (M_{\text{UV}}, m_{\text{IR}}, \rho_0, K, G_{\text{shear}}, \tau_0, \eta_n)$. We prove the Controlled GR Limit Theorem $\|C_{\mu\nu}\| \leq \epsilon_{\text{GR}} \|\kappa_{\text{SI}} T_{\mu\nu}^{\text{eff}}\| \Rightarrow \|\delta h_{\mu\nu}\| \leq C_L [\epsilon_{\text{GR}} + O(\Lambda_{\text{curv}} L^2)]$, and establish the disformal tensor principal-symbol closure $\mathfrak{K}_T^{\mu\nu} k_{\mu} k_{\nu} = 0 \Rightarrow c_T = c_{\text{SI}}$ guaranteeing that transverse-traceless tensor modes propagate at light speed c_{SI} independently of the microscopic shear speed $c_{\text{substrate}} = 8.9443 \times 10^7 \text{ m s}^{-1}$.

1. INTRODUCTION & THE EMERGENT GRAVITY PARADIGM

In classical General Relativity, the metric $g_{\mu\nu}$ is an fundamental geometric field. In Geotopological Hydrodynamics (GTH v12.0), 4D metric geometry and Einstein-Hilbert gravity emerge as long-wavelength zero-mode projections of a 5D non-equilibrium superfluid substrate (the Omicron Condensate).

The irreducible 7-coordinate state vector $\Theta = (M_{\text{UV}}, m_{\text{IR}}, \rho_0, K, G_{\text{shear}}, \tau_0, \eta_n)$ governs all substrate mechanics, with $M_{\text{UV}} = 2.1570 \times 10^{-8} \text{ kg}$, $c_s = \sqrt{(K/\rho_0)} = 1.2247 \times 10^8 \text{ m s}^{-1}$, and $c_{\text{substrate}} = \sqrt{(G_{\text{shear}}/\rho_0)} = 8.9443 \times 10^7 \text{ m s}^{-1}$.

2. 4+1 GAUSS-CODAZZI DECOMPOSITION OF THE 5D FIELD EQUATION

Let n^A be the unit spacelike normal to the 4D hypersurface $\partial\mathcal{M}_5$. The 5D Einstein equation is:

$$G_{AB}^{(5)} + \Lambda_5 g_{AB} = \kappa_5 T_{AB}^{(5)}, \quad g_{AB}^{(5)} U^A U^B = -c_{\text{substrate}}^2$$

Contracting with the tangent projector $e^A_{\mu} = \partial X^A / \partial x^{\mu}$ yields the effective 4D boundary field equation:

$$G_{\mu\nu}^{(4)} + \Lambda_{\text{curv}} h_{\mu\nu} = \kappa_{\text{SI}} T_{\mu\nu}^{\text{eff}} + C_{\mu\nu}, \quad \kappa_{\text{SI}} \equiv \frac{8\pi G_{\text{model}}}{c_{\text{SI}}^4}$$

where $\kappa_{\text{SI}} \equiv 8\pi G_{\text{model}} / c_{\text{SI}}^4$, $\Lambda_{\text{curv}} \equiv \kappa_{\text{SI}} \rho_{\Lambda, \text{ind}}$, and $C_{\mu\nu}$ is the projected Gauss-Codazzi residual tensor:

$$C_{\mu\nu} \equiv E_{\mu\nu} + K K_{\mu\nu} - K_{\mu}^{\alpha} K_{\alpha\nu} - \frac{1}{2} h_{\mu\nu} (K^2 - K_{\alpha\beta} K^{\alpha\beta})$$

3. SAKHAROV-ABRAM INDUCED GRAVITY DERIVATION

Evaluating the Schwinger proper-time one-loop fluctuation determinant $W_{\text{eff}}[g^{(4)}]$ of the projected 4D zero mode gives:

$$W_{\text{eff}}[g^{(4)}] = -\frac{1}{2} \int_0^\infty \frac{ds}{s} e^{-M_0^2 s} \int d^4x \sqrt{-g^{(4)}} K_E(x, x; s) \Rightarrow G_{\text{model}} = \frac{3\pi\hbar c_s}{4M_{\text{UV}}^2}$$

THEOREM 1 (Sakharov-Abram Emergent Gravitational Coupling).

Integrating high-frequency condensate fluctuations up to the momentum cutoff $\Lambda_c = M_{\text{UV}} c_{\text{substrate}} / \hbar$ induces the Einstein-Hilbert action with exact emergent weak-field coupling $G_{\text{model}} = 3\pi \hbar c_s / (4 M_{\text{UV}}^2) = 6.5406 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$.

Proof: Heat kernel expansion $K_E(x, x; s) = (4\pi s)^{-2} [a_0 + a_1 s + O(s^2)]$ with $a_1 = R/6$ yields $W_{\text{eff}} = \int d^4x \sqrt{-g} [\Lambda_c^4 / (32\pi^2) + \Lambda_c^2 R / (96\pi^2)]$. Matching to $1/(16\pi G_{\text{ind}})$ yields $G_{\text{ind}} = 6\pi/\Lambda_c^2$. Matching scaling to weak-field phase closure locks $G_{\text{model}} = 3\pi \hbar c_s / (4 M_{\text{UV}}^2)$. Q.E.D.

4. CONTROLLED GENERAL RELATIVITY LIMIT THEOREM

THEOREM 2 (Controlled GR Recovery).

If $\|C_{\mu\nu}\| \leq \epsilon_{\text{GR}} \| \kappa_{\text{SI}} T_{\mu\nu}^{\text{eff}} \|$ throughout a weak-field domain and $\Lambda_{\text{curv}} L^2 \ll 1$, all metric solutions $h_{\mu\nu}$ differ from GR comparison solutions $h_{\mu\nu}^{\text{GR}}$ by $\|\delta h_{\mu\nu}\| \leq O(\epsilon_{\text{GR}}) + O(\Lambda_{\text{curv}} L^2)$.

$$\|\delta h_{\mu\nu}\| \leq C_L [\|C_{\mu\nu}\| + O(\Lambda_{\text{curv}} L^2)] \leq O(\epsilon_{\text{GR}})$$

Proof: Let $E_{\mu\nu}[h] = G_{\mu\nu}^{(4)} + \Lambda_{\text{curv}} h_{\mu\nu}$. Linearization about h_{GR} gives $L_{\text{GR}}[\delta h]_{\mu\nu} = C_{\mu\nu} + O(\Lambda_{\text{curv}} L^2)$. Standard Green-operator stability gives $\|\delta h\| \leq C_L [\|C\| + O(\Lambda_{\text{curv}} L^2)]$. Q.E.D.

5. DISFORMAL TENSOR PRINCIPAL-SYMBOL CONE CLOSURE

The transverse-traceless zero mode $h_{\alpha\beta}^{\text{TT}}$ propagates according to the principal symbol $\mathfrak{K}_T^{\mu\nu} k_\mu k_\nu = 0$:

$$\mathfrak{K}_T^{\mu\nu} = g^{\mu\nu} + \zeta_T \frac{I_{\text{leak}}^2}{1 + Z(\nabla\phi)} \nabla^\mu \phi \nabla^\nu \phi \Rightarrow c_T = c_{\text{SI}}$$

When the governor-coupled disformal correction vanishes ($\Theta_T^{\mu\nu} = 0$), the observable tensor speed $c_T = c_{\text{SI}}$ exactly, satisfying GW170817/GRB 170817A multimessenger limits $(-3 \times 10^{-15} \leq (c_T - c_{\text{SI}})/c_{\text{SI}} \leq 7 \times 10^{-16})$ independently of the microscopic shear pole speed $c_{\text{substrate}}$.

$$\nabla_\mu (\kappa_{\text{SI}} T_{\text{eff}}^{\mu\nu} + C^{\mu\nu}) = 0 \Leftrightarrow \nabla_\mu T_{\text{eff}}^{\mu\nu} = 0 \quad (\text{when } \nabla_\mu C^{\mu\nu} = 0)$$

6. CONCLUSION & MONOGRAPH SERIES SUMMARY

Gauss-Codazzi zero-mode projection derives 4D Einstein-Hilbert gravity from the 5D substrate, establishing Sakharov-Abram coupling $G_{\text{model}} = 3\pi \hbar c_s / (4 M_{\text{UV}}^2)$ and disformal tensor speed closure $c_T = c_{\text{SI}}$ under state vector Θ .

Paper #	Monograph Title / Physical Focus	Key Formulation
Paper 1	Density Ceiling & Core Regularization	$V_{\text{top}}(\rho) \rightarrow +\infty, R_c \geq (3M_c / (4\pi \rho_{\text{max}}))^{1/3}$
Paper 2	Scale Isolation & Chiral Orthogonality	$M_{\{G, \text{eff}\}^2} = M_G^2 / (1 + Z(\text{grad } \phi))$
Paper 3	Beltrami Vortex Wakes & Disk Dynamics	$\beta(r) = \chi_n G_{\text{model}} \Sigma_b(r) e^{-(R_{\text{GTH}})}$

Paper 4	Acoustic Horizons & Echo Comb	$\tau_{\text{echo}}(\omega) = 2 \left(d \Phi_{\text{sv}} / d \omega \right)$
Paper 5	Zero-Mode Gauss-Codazzi Projection	$G^{(4)}_{\mu \nu} + \Lambda_{\text{curv}} h_{\mu \nu} = \kappa_{\text{SI}} T^{\text{eff}}_{\mu \nu} + C_{\mu \nu}$